



Grade 2 Science

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · GRADE 2 · FULL YEAR

160 min per week · 2 lessons weekly

About this overview. This document is an example of how the course might be taught. This overview is drawn from the Florida course description and Optima's 2026–27 course plan rather than a fully built course, so scholars may see some variation in investigations, examples, and projects from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Science Grade 2 invites scholars to practice the habits of young scientists: observing carefully, asking good questions, and using simple investigations to explore Earth and space, life science, and physical science. Scholars study weather, rocks and soil, and the Sun; plants, animals, the human body, and life cycles; and the properties of matter, motion, and energy. They build scientific vocabulary and communicate their discoveries through discussion, drawing, and short written explanations. Each week pairs one concept lesson with one hands-on investigation, and scholars keep a science notebook of their observations and drawings.

THE YEAR AT A GLANCE

1 The Nature of Science (body of knowledge)

Raising questions and investigating them, comparing observations made with the same tools, asking "how do you know?", telling observation from inference, and seeing why repeated investigations should agree.

2 Earth and Space Science (body of knowledge)

Rocks and how soil forms, classifying soil types, weather patterns and the seasons, the Sun's energy warming land, water, and air, evaporation, wind, and preparing for severe weather.

3 Physical Science (body of knowledge)

Measuring properties of objects; solids, liquids, and gases and water in its three states; altering materials; everyday uses of energy; and pushes, pulls, magnets, and how greater force changes motion.

4 Life Science (body of knowledge)

Human body parts and their functions, the life cycles of plants and animals such as beans and butterflies, the basic needs of living things, and how habitats meet those needs.

A TYPICAL WEEK

- Two science lessons each week.
- The first introduces a core idea through reading, examples, and guided observation.
- The second is hands-on: an investigation, sorting task, or simple experiment where scholars observe, describe, and record.
- Scholars keep a science notebook of their observations and labeled drawings.

TIME COMMITMENT

160 minutes / week

two lessons a week, about 80 minutes each

- Each week pairs one concept lesson with one hands-on investigation.
- Investigations and projects may run a little longer.

WHAT SCHOLARS NEED

- A science notebook for observations and labeled drawings.
- Simple household materials for hands-on investigations.
- A computer with reliable internet and access to Canvas.
- An adult to help young scientists with hands-on activities (especially in Kindergarten).

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting twice a week on the school's weekly schedule.